Certified Model Ranking under Limited Label Errors

Dev D. Goyal

***Abstract*—How many errors in shared test labels can a model ranking tolerate? Correcting one label can affect several comparisons because all models use the same examples. Test-label audits find errors in widely used benchmarks. Current evaluation methods include label audits, statistical confidence assessments, and comparisons between two systems. We study a worst-case question for fixed binary predictions. If only flagged labels may change, how much comparison weight can reverse under a given error budget? Checking comparisons individually can overstate the total reversal weight attainable under a single permitted correction. We construct finite lists that represent every permitted correction within the same error budget. Evaluating a list bounds the largest reversal weight from above and below. One tolerance shifts the accuracy margin used to count reversals. The other allows an additive error in reversal weight. For fixed tolerances, list size and construction cost are polynomial in the input length. Exact certification is coNP-complete even with a fixed margin and reversal-weight limit. A separate polynomial-time certificate tests which pairs of comparisons can reverse together at the requested margin. It gives a lower bound on the minimum number of label changes needed for the reversal weight to exceed the limit. A permitted correction whose reversal weight exceeds the limit gives an upper bound on that number. The tools distinguish a certified comparison rule, a permitted correction that violates it, and an unresolved result. Maintainers can report how many flagged label changes the stated rule provably tolerates.**

Impact Statement—Benchmark rankings guide model selection. Their conclusions depend on test labels, and audits show that some labels are wrong. Because all models use the same examples, correcting one label can change several comparisons at once. This paper gives a mathematical basis for reporting ranking sensitivity to label corrections in binary classification. A maintainer chooses comparison weights, an accuracy margin, and a reversal-weight limit. A certificate supports the rule under every permitted correction within the error budget. A permitted correction whose reversal weight exceeds the limit refutes the rule. When neither outcome is established, the audit remains unresolved. The methods also bound the minimum number of flagged label changes needed to violate the rule. These results concern fixed predictions and the specified labels and budget. They do not establish how often ranking changes occur on real benchmarks.

***Index Terms*—Model evaluation, label errors, ranking certificates, computational complexity, benchmarking, robustness certification.**

I. Introduction

A benchmark ranks AI models by accuracy on examples used by all models. Northcutt et al. estimate label errors in at least 6% of the ImageNet validation set [4]. MMLU reannotation also changes reported performance [5]. Correcting the labels of these examples can change several comparisons. We ask how much total comparison weight can reverse when only flagged labels may change under an error budget. Separate comparison tests can select incompatible corrections. They can therefore overstate the reversals that can occur together.[1]

The maintainer gives four inputs:

- The flagged labels and the maximum number that may change.
- A reference model order and weights for its comparisons.
- An accuracy margin that a later model's advantage must strictly exceed.
- A limit on the total weight of reversed comparisons.

The examples and predictions remain fixed. We consider every permitted correction without a probability distribution for the unknown labels.

With equal comparison weights and a reversal-weight limit of $1/10$, every permitted correction must leave at least 90 percent of comparisons without a reversal above the margin. Those comparisons can differ for each correction. The guarantee does not identify a common set of comparisons that survives every correction.

A finite representative list gives a lower bound by evaluation and an upper bound by coverage, with a margin shift and an additive reversal-weight tolerance. Every entry is within the error budget. A second method bounds simultaneous reversals at the requested margin. An upper bound can certify the rule. A permitted correction whose reversal weight exceeds the limit refutes it. A gap between these outcomes remains unresolved.

A. A comparison rule that tolerates one error

Take binary prediction rows $P_1 = (0, 0)$, $P_2 = (1, 0)$, and $P_3 = (1, 1)$ on two examples. The recorded labels are $(0, 0)$, and both may change. Use the displayed order, pair weights $1/3$, margin $t = 1/4$, and reversal-weight limit $\tau = 1/2$. A reversal requires a correct-count advantage of at least one example. Table I lists all corrections.

With one error, either adjacent pair can reverse, but they cannot reverse together (Figure 1). The maximum is $1/3$, below $1/2$. The per-comparison bound $2/3$ cannot certify the limit. With two errors, labels $(1, 1)$ reverse all three pairs. Thus the breakdown budget, the minimum number of changes needed to violate the limit, is two.

Dev D. Goyal is an independent researcher.

[1]AI assistance in this section: [1–3]. The Acknowledgment describes the scope of assistance.

TABLE I
TWO-EXAMPLE ILLUSTRATION WITH ALL FOUR CORRECTIONS. THE REFERENCE ORDER FOLLOWS DECREASING ACCURACY UNDER THE RECORDED LABELS. BOTH LABELS MAY CHANGE.

Labels	Changes	Correct counts	Reversed pairs	Reversed weight
00	0	$(2, 1, 0)$	none	0
10	1	$(1, 2, 1)$	$(1, 2)$	$1/3$
01	1	$(1, 0, 1)$	$(2, 3)$	$1/3$
11	2	$(0, 1, 2)$	all three	1

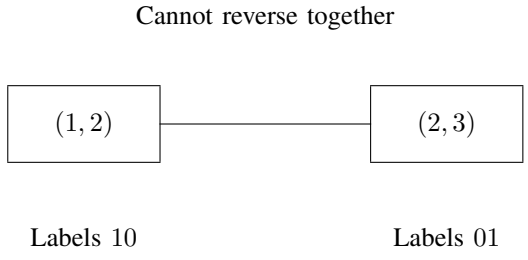


Fig. 1. Reversal events conflict under one error budget. The nodes are the possible reversals at $B = 1$ in Table I. The edge excludes their simultaneous occurrence. The per-comparison bound is $2/3$. The exact and conflict-aware values are $1/3$.

B. Obstacles and contributions

Coverage requires a nearby representative for each permitted correction. Each representative must be within the error budget. Removing over-budget representatives from an unrestricted cover can destroy coverage. The sample means used in our construction can have coordinates above one. We first find feasible fractional points and then round them to binary choices under the same budget.

The paper gives four connected results.

- Exact audit decisions are NP-complete or coNP-complete, according to the direction of the decision. Proposition 4 proves this with fixed margin and reversal-weight limit.
- A deterministic list construction with two-sided bounds on the largest reversal weight. Two complementary constructions give the sufficient bound in Theorem 1.
- A polynomial-time conflict-aware certificate at the requested margin. It tests whether two comparisons can reverse together and gives a verifiable upper bound in Theorem 3.
- Bounds on the breakdown budget. The conflict-aware certificate gives a lower bound. A permitted correction whose reversal weight exceeds the limit gives an upper bound.

For fixed comparison weights and an error budget, a single list supports every margin.

This is a theory paper. Its contributions are the problem formulation, complexity results, and verifiable certificates. Section V states the construction regimes and their limits. Section VII gives exact formulations for comparison.

The constructions use established spectral covering, sparse convex approximation, dependent rounding, and graph relaxations [6–10]. The covers here return feasible binary corrections under the same error budget. The proof gives the budget-dependent size and rational bit-operation bounds. A binary two-event test uses a linear scan instead of the general count-state dynamic program. The edge program and optional clique inequalities are established relaxations. Exact event tests turn their dual solutions into ranking certificates. The supplement gives the coverage records and dual checks needed for an upper bound.

Spectral covering and margin transfer do not by themselves give binary representatives within the correction budget. The feasible-cell construction in Supplement S2 and Proposition 2 supply that step. The sample means in our sparse convex approximation can have coordinates above one. Supplement S3 reconstructs feasible fractional points before the same rounding step. Dependent rounding preserves the budget; Proposition 2 also bounds the weighted pair-score error used here. The conflict relaxation is established. The exact event tests in Supplement S4 establish its edges from shared label corrections. These steps give the two budget-dependent bounds and the stated rational construction cost.

II. BACKGROUND AND RELATED WORK

a) Label quality and benchmark interpretation: Test-label audits show that scores depend on annotation choices. Northcutt et al.'s full-benchmark ImageNet comparison remains largely unchanged after identified errors are removed. Larger changes concern selected subsets or changes in noise prevalence [4]. Vasudevan et al. distinguish model mistakes from missing valid ImageNet labels [11]. MMLU re-annotation also changes reported performance [5]. These observed corrections do not bound every correction in an uncertainty set. Confident learning flags likely errors under a class-conditional noise model [12]. The flagged set remains an input to our certificate.[2]

b) Statistical comparison and leaderboard stability: Retrieval evaluation uses expected scores and pairwise confidence [13, Sec. 5]. Active testing estimates loss with bias correction [14]. Active comparison selects labels for two-model statistical tests [15]. Other work studies pipeline variation [16], item difficulty and model skill [17], or stability under changes to benchmark tasks and models [18]. Here predictions, examples, and comparison weights remain fixed while corrections to shared labels vary.

c) Robustness under changed data: Dataset multiplicity considers models trained on alternative datasets [19]. Programmable-bias certificates cover decision-tree predictions under specified training-data changes [20]. Regression audits study fractional sample-removal stability in low dimensions [21] and related exact and spectral methods [22]. These methods concern retraining or refitting. We change only the evaluation labels shared by fixed model predictions.

d) Bounds from incomplete information: Universal bounds include certain answers for incomplete databases [23], performance over compatible joint distributions [24], and possible-winner tests for partial preference orders [25]. Related

[2]AI assistance in this section: [1–3]. The Acknowledgment describes the scope of assistance.

evaluation work studies label acquisition for model choice in selective prediction [26], certifies reranking under accepted gain judgments [27], or maximizes retrieval-list effectiveness differences [28]. We bound weighted accuracy reversals under one shared correction budget.

III. The audit decision

Let $M \geq 2$ models make binary predictions on $N \geq 1$ examples. An $M \times N$ matrix gives their predictions, with one row per model and one column per example. Select $r \leq N$ flagged columns whose binary labels may change. All other labels remain fixed. Let $Z = \{0,1\}^r$ be the set of corrections of the flagged labels. Write $\hat{z}_u$ for the recorded label of flagged example u. The budgeted set is

$$Z_B = \{z \in Z : \textstyle\sum_u \mathbf{1}\{z_u \neq \hat{z}_u\} \leq B\}, \qquad B \in \{0, \ldots, r\}.$$

Thus Z_0 contains the unchanged labels and $Z_r = Z$. The error budget specifies the sensitivity analysis. It does not estimate the number of errors.[3]

Write $c_i(z)$ for model i's correct count and $y_i(z) = c_i(z)/N$. The denominator remains the original pool size. A reference order lists preferred models first. Assign rational comparison weights $\pi_e \geq 0$ of total one to pairs $e = (j, i)$ with j earlier than i. Define

$$\begin{aligned} d_e(z) &= y_i(z) - y_j(z), \\ C_{\pi,t}(z) &= \sum_e \pi_e \mathbf{1}\{d_e(z) > t\}, \\ U_B(t) &= \max_{z \in Z_B} C_{\pi,t}(z). \end{aligned} \tag{1}$$

The reversal weight $C_{\pi,t}(z)$ sums the weights of comparisons reversed by more than the margin. The audit asks whether $U_B(t) \leq \tau$ for given rational $t \geq 0$ and $0 \leq \tau < 1$. Equality at the margin does not count. With integer threshold $K = \lfloor Nt \rfloor + 1$, pair $e = (j, i)$ reverses exactly when $c_i(z) - c_j(z) \geq K$.

These inputs remain fixed when methods are compared. Uniform all-pairs weights measure the fraction of reversed claims. The top model's $M - 1$ comparisons carry weight $2/M$. A reversal-weight limit of at least $2/M$ can therefore permit all those comparisons to reverse.

One use of the aggregate rule is to assess a published set of comparison claims while allowing a stated fraction to fail. A maintainer may select the comparisons that the report is intended to support and set their weights before the audit. For equal weights and limit $1/10$, the question is whether every permitted correction leaves at least 90 percent of those claims without reversal above the margin. The surviving claims may differ between corrections. This evaluates the set as a whole. Protecting a particular winner is a different requirement, for which separate tests suffice.

At zero margin and uniform weights on all pairs, reversal weight is the fraction of strictly reversed pairs. If all corrected accuracies are distinct, it equals the normalized Kendall distance to the reference order. Corrected ties contribute zero here. This quantity is not a tie-adjusted rank statistic.

[3]AI assistance in this section: [1–3]. The Acknowledgment describes the scope of assistance.

TABLE II
Main inputs and outputs. Each limit controls a different quantity.

Symbol	Meaning
N, M	Number of examples and models.
r, B	Number of flagged labels and error budget.
π_e	Weight of comparison e. The weights sum to one.
t, τ	Accuracy margin and reversal-weight limit.
δ, ϵ	Margin-shift and reversal-weight tolerances.
$U_B(t)$	Largest reversal weight within the error budget.
$H_B(t)$	Conflict-aware upper bound at the requested margin.
b_L, b_*	Lower bound and exact value of the breakdown budget.

Flagging allows changes but does not prove errors. Weights can focus on selected models. The margin sets the required accuracy advantage. Reporting certified budgets over a range of B shows sensitivity to this input. Estimating B requires a validated label audit.

An upper bound at most τ certifies the rule for every permitted correction. A feasible correction $z \in Z_B$ with $C_{\pi,t}(z) > \tau$ refutes the rule. Otherwise the audit is unresolved. An upper bound above τ need not be attainable. All bounds must use the rule's flagged set, budget, order, weights, and margin.

For a nonempty feasible list $W \subseteq Z_B$, define $L_W(t) = \max_{w \in W} C_{\pi,t}(w)$. Direct evaluation gives $L_W(t) \leq U_B(t)$. An upper bound also requires coverage of every permitted correction. The next construction proves this coverage.

a) Scope and limitations: The guarantees apply to fixed binary predictions on the original examples. Changes are restricted to flagged labels under one budget. Budget validity requires separate evidence. The results do not establish population performance, guarantees for retraining or other metrics, practical runtime, annotation savings, or operational demand.

For later comparison, define the per-comparison upper bound under the same error budget:

$$B_{\mathrm{sep},B}(t) = \sum_e \pi_e \mathbf{1}\{\max_{z \in Z_B} d_e(z) > t\}, \quad U_B(t) \leq B_{\mathrm{sep},B}(t). \tag{2}$$

Each separate event is optimized under the same error budget B. The conflict-aware certificate checks whether two such events can occur together.

A single comparison can be checked by selecting the most helpful permitted flips. To protect one specified model against every rival, each rival must fail to overtake it under every correction. Separate tests suffice for this rule. Aggregating many possible reversals is harder. Exact certification is coNP-complete even at fixed margin and limit (Proposition 4). The approximation and upper certificates below address the aggregate decision.

IV. Constructing feasible correction lists

Each flagged label can flip, subject to the common error budget. We approximate the largest reversal weight with two tolerances. The tolerance δ allows a shift in the accuracy

margin. The tolerance ϵ allows an additive error in total reversal weight. Neither tolerance allows extra label changes.[4]

Theorem 1 (Combined binary correction bound)**.** *For rational $\delta, \epsilon \in (0,1)$, a deterministic algorithm constructs a nonempty $W \subseteq Z_B$ such that, for every real t,*

$$\begin{aligned} U_B(t+\delta) - \epsilon \le L_W(t) \le U_B(t) \\ \le L_W(t-\delta) + \epsilon. \end{aligned} \tag{3}$$

Its bit-operation count and list size are

$$2^{O\left(1+\frac{\min\{Br,\, B^2\log_2(r+1)\}}{\epsilon\delta^2 N^2}\right)}\,\mathrm{poly}(\ell), \tag{4}$$

where ℓ is the explicit rational input length. The polynomial degree is an absolute constant. The list can depend on B and the comparison weights, but not on t or the reversal-weight limit. Every output changes at most B labels.

One feasible list bounds the worst-case reversal weight from both sides at every margin, with margin shift δ and additive weight error ϵ. The upper bound evaluates the feasible list at a lower margin and then adds the reversal-weight tolerance. Corollary 1 gives polynomial construction for fixed tolerances. Bit complexity counts operations on binary encodings of inputs and intermediate rational numbers. The polynomial degree remains constant as the budget and tolerances change.

Corollary 1 (Polynomial construction at fixed tolerances)**.** *The list in Theorem 1 has size and construction cost*

$$2^{O(1+1/(\epsilon\delta^2))}\,\mathrm{poly}(\ell).$$

Thus both are polynomial in the explicit input length for every fixed pair of rational tolerances $\delta, \epsilon \in (0,1)$. The polynomial degree is an absolute constant.

Proof. Since $0 \le B \le r \le N$, we have $\min\{Br, B^2\log_2(r+1)\} \le Br \le N^2$. Substitute this inequality into (4). □

The two tolerances control different errors. This result does not give an additive approximation at the unchanged margin. It gives the simultaneous margin-shift bounds in (3). For the error fraction $\beta = B/N$ and flagged fraction $\phi = r/N$, the exponential parameter becomes

$$\frac{\min\{\beta\phi, \beta^2\log_2(r+1)\}}{\epsilon\delta^2}, \qquad 0 \le \beta \le \phi \le 1.$$

The recorded-label singleton applies when $\delta \ge 2\beta$.

The algorithm chooses a spectral cover with exponential parameter $Br/(\epsilon\delta^2N^2)$ or a sample-mean cover with parameter $B^2\log(r+1)/(\epsilon\delta^2N^2)$. Each bound is smaller in a different regime. Both constructions return feasible corrections.

A. The geometry and the common rounding step

Let x_u indicate a flip from the recorded label. A feasible vector uses only zero and one. During construction, we also use fractional coordinates between zero and one while keeping the same total error budget. Each correction z corresponds to the flip vector with $x_u = \mathbf{1}\{z_u \ne \hat{z}_u\}$. The feasible and fractional sets are

$$\begin{aligned} X_B &= \{x \in \{0,1\}^r : \textstyle\sum_u x_u \le B\}, \\ K_B &= \{x \in [0,1]^r : \textstyle\sum_u x_u \le B\}. \end{aligned}$$

Write $\beta_e = d_e(0)$ for the pair's accuracy difference under the recorded labels, and let a_{eu} be the change in its correct-count difference caused by flipping flagged label u. Then $d_e(x) = \beta_e + a_e^T x/N$, with $a_{eu} \in \{-2, 0, 2\}$. Set

$$\Sigma = \sum_e \pi_e a_e a_e^T, \qquad \|v\|_{\Sigma,N}^2 = v^T\Sigma v/N^2.$$

The matrix is positive semidefinite, with $\Sigma_{uu} \le 4$ and $\operatorname{tr}\Sigma \le 4r$. Its squared seminorm sums weighted squared score changes. A nonzero vector can have zero seminorm if it changes no positive-weight score. The construction does not use an inverse of Σ.

Write $W_B \subseteq X_B$ for the representative list in flip-vector notation; converting its vectors back to corrected labels gives $W \subseteq Z_B$. An unrestricted cover can assign $x \in X_B$ a representative $w \in X_r$ that exceeds the budget. Removing this representative can destroy coverage. Both constructions give $w \in W_B \subseteq X_B$ for every $x \in X_B$. Thus outputs and coverage use the same budget.

Both approximation branches use fractional points. The next rounding step preserves the budget and bounds squared score error.

Proposition 2 (Rounding under the exact error budget)**.** *For every rational $y \in K_B$, a deterministic polynomial-time rounding procedure returns $w \in X_B$ with*

$$\|w - y\|_{\Sigma,N}^2 \le 8B/N^2.$$

Proof. Add one dummy coordinate with zero coefficient so that the coordinate sum becomes the integer $\lceil \sum_u y_u \rceil \le B$. Extend Σ by a zero row and column. With y fixed, use

$$F(x) = (x-y)^T\Sigma(x-y) + 2\sum_u \Sigma_{uu}x_u(1-x_u).$$

Along a swap direction $e_i - e_j$ between two fractional coordinates, its quadratic coefficient is $-(e_i+e_j)^T\Sigma(e_i+e_j) \le 0$. Thus one endpoint of the feasible swap interval does not increase F. That endpoint makes at least one more coordinate integral and preserves the sum. The integer total prevents a single fractional coordinate from remaining. Repeating this step gives an integral vector. Its distance term is at most $F(y) \le 8\sum_u y_u \le 8B$, because the dummy has zero diagonal. Delete the dummy and divide by N^2. For the bit bound, let Q be the product of the denominators of the input coordinates of y. Its bit length is at most their total input length. The dummy and all initial coordinates are multiples of $1/Q$. Each swap endpoint moves a coordinate to zero or one and changes the other by a multiple of $1/Q$. Thus all iterates have the same denominator and stay in $[0,1]$. There are at most $r+1$ swaps. Exact evaluations of F use a polynomial number of rational operations on these coordinates and the fixed rational matrix Σ. Their bit lengths are polynomial in the input length. The procedure therefore uses a polynomial number of bit operations. □

[4] AI assistance in this section: [1–3]. The Acknowledgment describes the scope of assistance.

Swapping mass between fractional coordinates preserves their total and bounds score error. This is a specialization of dependent rounding and swap concavity [9, 29–31]. Independent rounding alone would not preserve the budget.

Choose the least integer $j \geq 0$ such that $4^{-j} \leq \epsilon$, and put $\nu = \delta 2^{-j}$. Then

$$\epsilon\delta^2/4 < \nu^2 \leq \epsilon\delta^2.$$

For the nontrivial cover branches, it is enough to approximate every $x \in X_B$ within ν in the score seminorm.

B. Spectral construction

The spectral construction retains a subspace that contains large score variations. A finite grid covers this subspace. The omitted directions contribute a bounded error. The squared Euclidean radius of K_B is at most B. The retained-coordinate argument uses the scale $B \operatorname{tr}\Sigma/N^2 \leq 4Br/N^2$. For a sufficiently large pool, the grid gives feasible fractional representatives. Proposition 2 then gives feasible outputs. Otherwise the stated exponent permits enumeration of the full binary cube and retention of the feasible outputs.

The resulting size and bit-operation count are

$$2^{O(1+Br/(\epsilon\delta^2N^2))}\operatorname{poly}(\ell).$$

Supplement S2 proves the rational spectral grid, the search for feasible representatives, bounds on the number of partial grid assignments examined, and small-pool inequalities. Covariance truncation and squared-error margin transfer follow the method in Diakonikolas et al. [6, Sec. 2.1, Lemma 2.2 and Algorithm 1]. The rational ellipsoid traversal, feasible representatives, and bit bounds are proved here.

C. Sample-mean construction

The second construction uses averages of simple vectors to approximate the pair scores. Assume $B \geq 1$. For any $x \in K_B$, let Y equal Be_u with probability x_u/B, and zero otherwise. Here e_u is the uth coordinate unit vector. Then $\mathbb{E}Y = x$ and

$$\mathbb{E}\|Y - x\|^2_{\Sigma,N} \leq 4B^2/N^2.$$

The mean of $k = \lceil 64B^2/(\nu^2N^2)\rceil$ independent copies has expected squared error at most $\nu^2/16$. A word is an ordered sequence of these simple vectors. At least one word from $\{0, Be_1, \ldots, Be_r\}^k$ has mean c within $\nu/4$ of x. The algorithm enumerates all words, so it is deterministic. Randomness proves the existence of a good word. It does not describe uncertainty in the test labels. This uses the established sampling argument for approximate convex representation [7].

A mean can have a coordinate above one. For each mean, a conditional-gradient procedure using exact rational arithmetic [8] finds a feasible point $y \in K_B$. Each step combines a feasible point with a solution of a linear optimization over the feasible set. The procedure minimizes the quadratic $\|y - c\|^2_{\Sigma,N}$ to additive error $\nu^2/16$. For a mean that covers x, the feasible point x bounds the optimum by $\nu^2/16$. Thus $\|y - c\|_{\Sigma,N} \leq \nu/\sqrt{8}$. If $N^2 \geq 128B/\nu^2$, Proposition 2 gives a feasible w with rounding error at most $\nu/4$. The approximation, feasibility, and rounding errors give

$$\|x - w\|_{\Sigma,N} \leq \nu/4 + \nu/\sqrt{8} + \nu/4 < \nu.$$

Otherwise $k > B/2$, and direct feasible-subset enumeration has size at most $(r+1)^B \leq (r+1)^{2k}$. Supplement S3 proves the exact oracle, convergence, iteration count, and rational bit bounds. In both branches the size and bit-operation count are

$$(r+1)^{O(1+B^2/(\epsilon\delta^2N^2))}\operatorname{poly}(\ell). \tag{5}$$

The factor $(r+1)^{O(1)}$ is polynomial in the explicit input length and can be absorbed into $\operatorname{poly}(\ell)$. Its degree is an absolute constant.

D. Choosing a bound and transferring thresholds

If $r = 0$ or $B = 0$, return the recorded-label correction. The same rule applies if $\Sigma = 0$, because all positive-weight pair scores are then constant. If $\delta \geq 2B/N$, the recorded-label singleton gives the two-sided bound with zero reversal-weight error. Each pair score can change by at most $2B/N$. This branch gives the threshold guarantee directly. It need not give a seminorm cover of radius ν.

Otherwise, $\delta < 2B/N$, so $q = B^2/(\epsilon\delta^2N^2) > 1/(4\epsilon) > 1/4$. Thus the additive one in the exponent of (5) can be absorbed into q. Let $L = \lceil \log_2(r+1)\rceil$, found by integer comparisons. Choose the spectral construction if $r \leq BL$, and the sample construction otherwise. Write $l = \log_2(r+1)$, so $l \leq L \leq 2l$. If the rule selects the spectral branch and $r > Bl$, then $Br \leq B^2L \leq 2B^2l$. If it selects the sample branch, then $r > BL \geq Bl$, so that branch has the smaller parameter. The other spectral case also selects the smaller parameter. Thus the rule loses at most a factor two. It proves (4) using exact integer comparisons.

In either nontrivial cover branch, every x has a feasible representative w with weighted squared score error at most ν^2. By Markov's inequality, the comparisons for which $|d_e(x) - d_e(w)| > \delta$ have total weight at most $\nu^2/\delta^2 \leq \epsilon$. This converts weighted squared score error into reversal-weight error. For every other comparison,

$$d_e(x) > t + \delta \implies d_e(w) > t,$$
$$d_e(x) > t \implies d_e(w) > t - \delta.$$

Both implications preserve strict events, including equality in the error bound. They give $C_{\pi,t+\delta}(x) \leq C_{\pi,t}(w) + \epsilon$ and $C_{\pi,t}(x) \leq C_{\pi,t-\delta}(w) + \epsilon$. The same w works for every real t. Maximizing over x and using output feasibility proves (3). In the recorded-label branch, all errors are at most δ. The same implications therefore apply with no exceptional weight. This completes the proof from the two cover constructions.

E. Verifying a claimed upper bound

A claimed upper bound requires complete coverage evidence. Check every spectral traversal and representative, every sample word and its exact updates and output association, or the complete fallback enumeration. The recorded-label branch requires its threshold condition. Duplicate outputs may share

storage while retaining all associations. Supplement S5 gives the complete checks and error bounds.

At the requested margin, $L_W(t-\delta)+\epsilon \le \tau$ certifies the audit. A listed correction with $C_{\pi,t}(w) > \tau$ refutes it. A gap between these tests is unresolved. The upper bound does not allow extra label errors. The margin shift can also limit the certificate. Comparisons close to the requested margin may contribute to $L_W(t-\delta)$ even when they do not contribute at t. Thus a valid upper bound can remain too large to certify the rule.

V. Construction regimes

a) When approximation replaces enumeration: Put $\eta = 4^{-j}$, where j is the least nonnegative integer satisfying $4^{-j} \le \epsilon$. Then $\nu^2 = \eta\delta^2$ and $\epsilon/4 < \eta \le \epsilon$. After the zero and recorded-label cases, either construction uses its approximation branch exactly when

$$\eta\delta^2 N^2 \ge 128B.$$

Otherwise it uses its exact enumeration fallback. Outside the recorded-label case, $\delta N < 2B$. Together these two inequalities require

$$B > 32/\eta \ge 32/\epsilon.$$

This budget condition is necessary, but it does not replace the pool-size condition. For example, $\epsilon = 1/4$ requires $B \ge 129$. For $\epsilon = 1/20$, the chosen $\eta = 1/64$ requires $B \ge 2049$. These thresholds describe the stated construction constants. They are not lower bounds on other algorithms.

For example, suppose $\Sigma \ne 0$ and take $N = 10{,}000$, $r = 1{,}000$, $B = 100$, $\delta = 0.01$, and $\epsilon = 0.05$. The recorded-label branch requires $\delta \ge 0.02$ and does not apply. Here $\eta = 1/64$, so $\eta\delta^2N^2 = 156.25 < 128B = 12{,}800$. The branch rule selects the spectral construction because $\lceil \log_2(r+1) \rceil = 10$ and $r \le 10B$. The stated construction therefore falls back to scanning the 2^{1000} binary cube. This is a consequence of the construction, not an observed runtime or a lower bound on other methods. The list theorem establishes an approximation-complexity result. It does not establish a practical list algorithm for these parameters.[5]

For such parameters, the conflict-aware certificate in Section VI and the exact formulation in Section VII-A provide other audit methods. The binary event tests take $O(r+B+1)$ operations after the recorded count differences are known. Their practical runtime remains to be measured.

Table III compares sufficient bounds, whose minimum in (4) need not be optimal. Setting $B = r$ gives the full-budget spectral scale $r^2/(\epsilon\delta^2N^2)$.

For error fraction $\beta = B/N$ and flagged fraction $\phi = r/N$, the two exponential parameters are $\beta\phi/(\epsilon\delta^2)$ and $\beta^2 \log_2(r+1)/(\epsilon\delta^2)$. The integer selection rule compares these bounds. Supplement S6 gives two explicit families: one has a polynomial sample-mean bound and an exponential spectral bound; the other has the smaller spectral bound. These are comparisons of sufficient bounds, not measured times or lower bounds on the best algorithm.

[5]AI assistance in this section: [1–3]. The Acknowledgment describes the scope of assistance.

TABLE III
Budgeted binary lists. Put $D_0 = \epsilon\delta^2N^2$. Fixed polynomial factors are omitted in the cover rows.

Construction	Sufficient size and condition
Recorded-label singleton	One; zero reversal-weight error if $\delta \ge 2B/N$.
Exact feasible enumeration	$\sum_{j=0}^{B} \binom{r}{j}$; exact at every margin.
Spectral cover	$2^{O(1+Br/D_0)}$; margin shift δ, reversal-weight error ϵ.
Sample-mean cover	$(r+1)^{O(1+B^2/D_0)}$; the same tolerances and exact error budget.
Combined construction	(4); includes the recorded-label branch.

VI. A conflict-aware certificate at the requested margin

The conflict-aware bound retains the requested margin. Neither it nor the list bound is proved to dominate the other. Let V_B contain positive-weight pairs with possible events $R_e = \{z \in Z_B : d_e(z) > t\}$. The graph $G_{B,t}$ joins events that cannot occur together. Define

$$\begin{aligned} H_B(t) = \max \sum_{e \in V_B} \pi_e x_e \\ \text{s.t. } 0 \le x_e \le 1 \quad (e \in V_B), \\ x_e + x_f \le 1 \quad (ef \in E(G_{B,t})). \end{aligned} \tag{6}$$

The program allows fractional values between zero and one. Each edge excludes both endpoints taking value one. Every correction gives a feasible zero-one event vector. Some fractional vectors have no realizing correction. The optimum is therefore an upper bound. This standard edge relaxation for weighted vertex packing [10, 32] uses linear and dual bounds also found in related optimization methods [33].[6]

Theorem 3 (Budgeted conflict-aware certificate)**.** *The graph and $H_B(t)$ are computable deterministically with a polynomial number of bit operations. They satisfy*

$$U_B(t) \le H_B(t) \le B_{\mathrm{sep},B}(t). \tag{7}$$

An optimal rational dual certificate exists with at most $|V_B|$ positive entries and polynomial total bit length. For fixed predictions, flagged labels, recorded labels, order, comparison weights, and margin, $H_b(t) \le H_{b+1}(t)$ and $H_0(t) = U_0(t)$.

Pairwise conflict checks therefore never weaken the per-comparison upper bound. They give a polynomial-time bound with a rational dual certificate.

For binary labels, write $\Delta_e(0) = N\beta_e$ for the recorded correct-count difference of pair e; the zero argument denotes no label changes. A one-event test needs one scan: if n_+ flagged flips increase this difference by two, its maximum under budget B is $\Delta_e(0) + 2\min\{B, n_+\}$. Two events can also be tested in $O(r+B)$ integer arithmetic operations after their recorded count differences are known. The test groups flip gains into nine possible pairs. It includes shared positive gains and checks the remaining tradeoffs. Supplement S4

[6]AI assistance in this section: [1–3]. The Acknowledgment describes the scope of assistance.

gives the exact test and proof. Thus graph construction uses $O(P^2(r + B + 1))$ arithmetic operations after the recorded gaps, for P positive-weight comparisons. The bit lengths are polynomial in the input length.

For more general candidate-label domains, the event tests process one example at a time. A one-event or two-event dynamic program records the integer count differences and the number of changed labels. Each state gives the results still possible after the processed examples. The program retains only states within the error budget. With D the sum of explicitly listed permitted domain sizes ($D \le 2N$ in the binary case), a two-event test takes $O(N^2(B + 1)D)$ transitions. With $P \le \binom{M}{2}$ positive-weight pairs, graph construction uses $O(P^2)$ such tests, in addition to single-event tests and a rational linear-program solve. Supplement S4 proves the state correspondence, bit bounds, dual, and monotonicity.

A dual solution gives an upper bound through inequalities that can be checked directly. The dual covers comparison weights with nonnegative vertex amounts $\alpha_e \ge 0$ and edge amounts $\beta_{ef} \ge 0$. Require

$$\alpha_e + \sum_{f: ef \in E(G_{B,t})} \beta_{ef} \ge \pi_e, \quad \sum_e \alpha_e + \sum_{ef} \beta_{ef} \le u_{\mathrm{up}}. \quad (8)$$

Then $U_B(t) \le u_{\mathrm{up}}$ by weak duality. A checker verifies that omitted positive-weight events are impossible, checks edges with positive β_{ef}, and checks (8) exactly. Unused edges need not be checked for the claimed upper bound. To establish the exact value $H_B(t)$, also provide a primal solution feasible for the full conflict graph $G_{B,t}$ whose objective equals the dual objective, or solve the linear program exactly.

A third party can verify a published certified tolerance by checking its event tests and rational dual bound, without repeating the optimization search.

Define the breakdown budget and its certified lower bound by

$$b_* = \min\{b \in \{0, \dots, r\} : U_b(t) > \tau\},$$
$$b_L = \min\{b \in \{0, \dots, r\} : H_b(t) > \tau\}.$$

Set an empty-set minimum to $r+1$. Then $b_L \le b_*$. All budgets below b_L are certified. Hence the certified error tolerance is $b_L - 1$ when $b_L \ge 1$. If $b_L = 0$, the recorded labels already exceed the limit; if $b_L = r + 1$, all r flagged changes are certified. A permitted correction with c changes whose reversal weight exceeds the limit gives $b_* \le c$. Thus an upper certificate and such a correction bound the breakdown budget from opposite sides. An upper bound above the limit alone does not prove breakdown.

In Table I, $H_0 = 0$, $H_1 = 1/3$, and $H_2 = 1$. At $B = 1$, comparison $(1, 3)$ cannot reverse. Dual weight $1/3$ on the adjacent events' conflict edge certifies $1/3$. Either one-change correction attains this value. At $B = 2$, labels 11 give reversal weight one. Thus $b_L = b_* = 2$ at limit $1/2$.

The same example gives a list certificate. Use $W = Z_1$, $\delta = 1/8$, and $\epsilon = 1/12$. Then $L_W(1/8) + \epsilon = 5/12 < 1/2$, which certifies the original margin $1/4$. Exact enumeration gives the stronger value $1/3$. The calculation illustrates the decision rule. The recorded-label singleton fails the two-sided bound because $U_1(3/8) = 1/3 > \epsilon$.

TABLE IV
COSTS OF THE AUDIT QUESTIONS. SCAN COSTS ASSUME RECORDED CORRECT COUNTS, COMPUTED IN $O(MN)$ OPERATIONS, AND THE REQUIRED RECORDED COUNT DIFFERENCES. GENERAL POLYNOMIAL BOUNDS COUNT BIT OPERATIONS.

Question	Cost
Can one comparison reverse?	$O(r + 1)$
Can any rival overtake one model?	$O(M(r + 1))$
Can two comparisons reverse together?	$O(r + B + 1)$ for binary labels
Exact $U_B(t)$ for fixed M	Polynomial; degree depends on M
Exact certification in general	coNP-complete, Prop. 4
Two-sided list bound	Polynomial at fixed δ, ϵ, Cor. 1
Conflict-aware bound $H_B(t)$	Polynomial, Thm. 3
Lower breakdown bound b_L	$O(\log(r+2))$ bound evaluations

The edge relaxation limits how much the conflict-aware bound can improve on per-comparison testing. Since $x_e = 1/2$ is feasible in (6), $H_B \ge B_{\mathrm{sep},B}/2$. Table I attains equality at $B = 1$. This lower bound on H_B compares two upper certificates. It does not compare either certificate with the exact value U_B. The edge certificate can improve the decision only in the range $\tau < B_{\mathrm{sep},B} \le 2\tau$. This is a necessary condition, not a sufficient one. A supplied clique, a set of pairwise conflicting events, permits the stronger constraint $\sum_{e \in C} x_e \le 1$. Any supplied family of polynomial total size gives a polynomial-size augmented program and a checkable dual. Supplement S4 states the verification conditions. Such constraints can exceed the edge relaxation's factor-two improvement over the per-comparison bound. They still cannot detect a restriction with no pairwise conflict. Supplement S6 gives an example with $H_B = 1$ and $U_B = 3/4$.

A. Reporting a certified error tolerance

A maintainer first flags suspect labels, for example with confident learning [12], and fixes the reference order, comparison weights, margin t, and limit τ. For a trial budget B, run the one-event test on each positive-weight comparison. A reversal requires the earlier model's recorded accuracy lead to be below $2B/N - t$. With $N = 10{,}000$ and $B = 50$, only leads below $0.01 - t$ can reverse. Then build V_B and the conflict graph and solve (6). Because $H_B(t)$ is nondecreasing in B, binary search finds b_L with $O(\log(r + 2))$ bound evaluations. Each evaluation builds the graph for its trial budget. Report b_L as the lower bound on the breakdown budget and $b_L - 1$ as the certified error tolerance when $b_L \ge 1$, with the endpoint cases stated above. A violating correction from (9) supplies an upper bound on b_*, even if the solver stops before proving optimality.

VII. EXACT COMPUTATION AND SCOPE EXTENSIONS

Table IV summarizes the costs of the audit questions.

Proposition 4 (Exact audit complexity)**.** *Deciding whether $U_B(t) > \tau$ is NP-complete. Deciding whether $U_B(t) \le \tau$ is coNP-complete. Both statements hold with $B = r = N$, $t = \tau = 1/2$, and equal positive weights on a matching of model rows. Every comparison outside that matching has*

weight zero. The reference order can sort the models by non-increasing accuracy under the recorded labels. The number of models is part of the input, and repeated prediction rows are permitted.

Unless P equals NP, no polynomial-time algorithm decides every such audit exactly. This worst-case limit does not rule out efficient exact solutions on particular instances.

The reduction is from binary Closest String [34]. A designated model pair reverses exactly when a corrected string lies within a prescribed Hamming radius. Padding fixes the accuracy margin. One additional event and never-reversing pairs fix the reversal-weight limit. Supplement S6 proves the reduction and the recorded-accuracy order. The result concerns exact decisions. Its score and weight gaps shrink with the input size, so it is compatible with the fixed-tolerance corollary. It does not prove hardness for equal positive weights on all model pairs.[7]

A. An exact integer formulation

Let $E_+ = \{e : \pi_e > 0\}$, let $\Delta_e(x) = N\beta_e + \sum_{u=1}^{r} a_{eu}x_u$, and set $K = \lfloor Nt \rfloor + 1$. The count difference $\Delta_e(x)$ is an integer. Let $\underline{\Delta}_e$ be its minimum on X_B. It is obtained by adding to $N\beta_e$ the at most B most negative coefficients a_{eu}. Put $M_e = \max\{0, K - \underline{\Delta}_e\}$. The following mixed-integer linear program has optimum $U_B(t)$:

$$\begin{aligned} \max \quad & \sum_{e \in E_+} \pi_e s_e \\ \text{subject to} \quad & \sum_{u=1}^{r} x_u \le B, \qquad x_u \in \{0,1\}, \\ & s_e \in \{0,1\}, \\ & \Delta_e(x) \ge K - M_e(1 - s_e) \quad (e \in E_+). \end{aligned} \tag{9}$$

When $s_e = 0$, the linking inequality holds for every $x \in X_B$. When $s_e = 1$, it requires the reversal event. Thus any feasible objective is at most the reversal weight of its correction. Conversely, every correction admits its actual event indicators as the values of s_e. Maximizing proves the identity. An event indicator can be zero at a feasible point even when the event holds. This does not affect the optimum because all objective weights are positive.

For binary labels, group the flagged columns by their recorded correctness vector $v \in \{0,1\}^M$, where $v_m = 1$ if model m agrees with that column's recorded label and $v_m = 0$ otherwise. Let m_v be the number of flagged columns of type v, and use an integer variable $n_v \in \{0, \dots, m_v\}$ for the number changed. Replace $\sum_u x_u \le B$ by $\sum_v n_v \le B$ and replace the score contribution by

$$\Delta_{(j,i)}(n) = N\beta_{(j,i)} + 2\sum_{v}(v_j - v_i)n_v.$$

There are at most $\min\{r, 2^M\}$ types. Every correction gives these counts, and every feasible count vector can be realized by selecting the stated number of columns in each type. The aggregated formulation therefore has the same optimum. This exact formulation supplies a comparison method. Its solution time is not bounded by a polynomial here.

For fixed M, a dynamic program can record all M correct counts and the change count after each example. There are at most $(N+1)^M(B+1)$ states per layer. The final states determine $U_B(t)$. Thus fixed model count permits polynomial exact computation, with an exponent that depends on M.

The conflict-aware certificate has a polynomial construction guarantee and explicit exact checks. The integer formulation remains useful for exact values when optimality is proved and for violating corrections found before termination.

B. One candidate replacement per example

The list construction also covers multiclass predictions when each flagged example has one fixed alternative to its recorded label. A binary variable specifies whether to change that label. The pair coefficients can now be $-2, -1, 0, 1, 2$, and their absolute bound remains two. The matrix bounds, rounding, and both covers are unchanged. Supplement S6 gives the full argument. The conflict dynamic program permits any explicitly listed finite candidate set. This broader statement does not extend the list construction to several alternatives per example.

C. Exploratory implementation check

An exploratory implementation check used 20 synthetic binary tables ($N = 200$, $M = 6$, $r = 12$, $t = 0$). Independent exhaustive checks found no event-test mismatch. Exact rational checks verified all 260 graph certificates. At $\tau = 1/10$ and $1/5$, the breakdown lower bound b_L exceeded the bound from separate tests in 5/20 and 10/20 instances. The exact breakdown was finite in 16 and 5 instances, respectively. Among those finite cases, $b_L = b_*$ in 16/16 and 3/5 instances. The remaining 4 and 15 instances, respectively, had no violation through r; in each, $b_L = r + 1$. Supplement S7 gives the methods, full counts, and limitations.

VIII. Conclusion and Future Work

Shared labels can make separate worst-case reversals incompatible. Representative lists account for this dependence while keeping the error budget, with a margin shift and additive reversal-weight tolerance. A conflict-aware certificate excludes incompatible event pairs at the requested margin. It gives a lower bound on the breakdown budget. A permitted correction whose reversal weight exceeds the limit gives an upper bound. Both bounds equal two in the two-example illustration. The guarantee concerns weighted comparison rules. It does not require every model to keep its position.[8]

Exact decisions remain hard even with fixed margin and reversal-weight limit. Fixed approximation tolerances permit polynomial construction. This distinction separates the computational barrier from the cost of the stated approximate guarantee. The bounds do not give a tight complexity characterization. The hardness construction places positive weight

[7] AI assistance in this section: [1–3]. The Acknowledgment describes the scope of assistance.

[8] AI assistance in this section: [1–3]. The Acknowledgment describes the scope of assistance.

only on disjoint model pairs, and its score and reversal-weight gaps shrink with the input size. It does not give a matching lower bound on the tolerance dependence in Theorem 1.

The next questions follow from the current limits. An implementation should measure when list construction and conflict checking improve on exact enumeration. A validated label audit should test how to choose the flagged set and error budget. Stronger constraints involving several reversal events could reduce the gap left by pairwise conflict tests. Further theory is needed for several candidate replacement labels in the list construction, unequal correction costs, several budgets, and sharp dependence on the tolerances.

The supplement gives complete proofs and rational construction details. The small synthetic implementation check covers the conflict-aware certificate. The list constructions are not evaluated empirically. Measuring how often shared corrections change certification decisions in benchmark prediction tables, and whether the certificate saves computation over exact solvers, remains an open empirical question.

ACKNOWLEDGMENT

OpenAI Codex, with GPT-6 Sol, GPT-6 Astra, and GPT-6.1 Sol, assisted with proof development, mathematical and scientific checks, drafting, and revision. OpenAI assistance with the synthetic implementation check covered code design, execution, verification, and results wording. The affected text includes the abstract, impact statement, all main sections, and all supplement sections. ChatGPT suggested the sample-mean comparison and the two-example illustration. Anthropic Claude Opus 5.5 also provided scientific assistance with the mathematical arguments and manuscript. The assistance included mathematical content and was not limited to language editing. The author is responsible for the claims and text.

REFERENCES

[1] OpenAI, "Codex," 2026, AI assistance; accessed October 30, 2026. [Online]. Available: https://openai.com/codex/

[2] ——, "ChatGPT," 2026, AI assistance; accessed October 30, 2026. [Online]. Available: https://chatgpt.com/overview/

[3] Anthropic, "Claude," 2026, AI assistance; accessed October 30, 2026. [Online]. Available: https://claude.com/product/overview

[4] C. Northcutt, A. Athalye, and J. Mueller, "Pervasive label errors in test sets destabilize machine learning benchmarks," in *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, vol. 1, 2021. [Online]. Available: https://datasets-benchmarks-proceedings.neurips.cc/paper/2021/hash/f2217062e9a397a1dca429e7d70bc6ca-Abstract-round1.html

[5] A. P. Gema, J. O. J. Leang, G. Hong, A. Devoto, A. C. M. Mancino, R. Saxena, X. He, Y. Zhao, X. Du, M. R. Ghasemi Madani, C. Barale, R. McHardy, J. Harris, J. Kaddour, E. Van Krieken, and P. Minervini, "Are we done with MMLU?" in *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. Association for Computational Linguistics, 2025, pp. 5069–5096.

[6] I. Diakonikolas, D. M. Kane, and P. Manurangsi, "Nearly tight bounds for robust proper learning of halfspaces with a margin," in *Advances in Neural Information Processing Systems*, vol. 32, 2019, full version, arXiv:1908.11335v1. [Online]. Available: https://arxiv.org/abs/1908.11335v1

[7] V. Mirrokni, R. Paes Leme, A. Vladu, and S. C.-w. Wong, "Tight bounds for approximate Carathéodory and beyond," in *Proceedings of the 34th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 70, 2017, pp. 2440–2448. [Online]. Available: https://proceedings.mlr.press/v70/mirrokni17a.html

[8] M. Jaggi, "Revisiting Frank–Wolfe: Projection-free sparse convex optimization," in *Proceedings of the 30th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 28, 2013, pp. 427–435, part 1. [Online]. Available: https://proceedings.mlr.press/v28/jaggi13.html

[9] R. Gandhi, S. Khuller, S. Parthasarathy, and A. Srinivasan, "Dependent rounding and its applications to approximation algorithms," *Journal of the ACM*, vol. 53, no. 3, pp. 324–360, May 2006.

[10] A. Atamtürk, G. L. Nemhauser, and M. W. P. Savelsbergh, "Conflict graphs in solving integer programming problems," *European Journal of Operational Research*, vol. 121, no. 1, pp. 40–55, 2000.

[11] V. Vasudevan, B. Caine, R. Gontijo Lopes, S. Fridovich-Keil, and R. Roelofs, "When does dough become a bagel? analyzing the remaining mistakes on ImageNet," in *Advances in Neural Information Processing Systems*, vol. 35. Curran Associates, Inc., 2022, pp. 6720–6734.

[12] C. Northcutt, L. Jiang, and I. Chuang, "Confident learning: Estimating uncertainty in dataset labels," *Journal of Artificial Intelligence Research*, vol. 70, pp. 1373–1411, 2021.

[13] B. Carterette, J. Allan, and R. Sitaraman, "Minimal test collections for retrieval evaluation," in *Proceedings of the 29th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*. ACM, 2006, pp. 268–275.

[14] J. Kossen, S. Farquhar, Y. Gal, and T. Rainforth, "Active testing: Sample-efficient model evaluation," in *Proceedings of the 38th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 139, 2021, pp. 5753–5763. [Online]. Available: https://proceedings.mlr.press/v139/kossen21a.html

[15] C. Sawade, N. Landwehr, and T. Scheffer, "Active comparison of prediction models," in *Advances in Neural Information Processing Systems*, vol. 25, 2012. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2012/file/92fb0c6d1758261f10d052e6e2c1123c-Paper.pdf

[16] X. Bouthillier, P. Delaunay, M. Bronzi, A. Trofimov, B. Nichyporuk, J. Szeto, N. Mohammadi Sepahvand, E. Raff, K. Madan, V. Voleti, S. Ebrahimi Kahou, V. Michalski, T. Arbel, C. Pal, G. Varoquaux, and P. Vincent, "Accounting for variance in machine learning benchmarks," in *Proceedings of Machine Learning and Systems*, vol. 3, 2021, pp. 747–769. [Online]. Available: https://proceedings.mlsys.org/paper_files/paper/2021/hash/0184b0cd3cfb185989f858a1d9f5c1eb-Abstract.html

[17] P. Rodriguez, J. Barrow, A. Hoyle, J. P. Lalor, R. Jia, and J. Boyd-Graber, "Evaluation examples are not equally informative: How should that change NLP leaderboards?" in *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*. Association for Computational Linguistics, 2021, pp. 4486–4503.

[18] G. Zhang and M. Hardt, "Inherent trade-offs between diversity and stability in multi-task benchmarks," in *Proceedings of the 41st International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 235. PMLR, 2024, pp. 58 984–59 002. [Online]. Available: https://proceedings.mlr.press/v235/zhang24u.html

[19] A. P. Meyer, A. Albarghouthi, and L. D'Antoni, "The dataset multiplicity problem: How unreliable data impacts predictions," in *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency*. Association for Computing Machinery, 2023, pp. 193–204.

[20] A. Meyer, A. Albarghouthi, and L. D'Antoni, "Certifying robustness to programmable data bias in decision trees," in *Advances in Neural Information Processing Systems*, vol. 34. Curran Associates, Inc., 2021, pp. 26 276–26 288. [Online]. Available: https://proceedings.neurips.cc/paper/2021/hash/dcf531edc9b229acfe0f4b87e1e278dd-Abstract.html

[21] A. Moitra and D. Rohatgi, "Provably auditing ordinary least squares in low dimensions," in *The Eleventh International Conference on Learning Representations*, 2023. [Online]. Available: https://openreview.net/forum?id=DlpCotqdTy

[22] D. Freund and S. B. Hopkins, "Towards practical robustness auditing for linear regression," 2023, arXiv:2307.16315. [Online]. Available: https://arxiv.org/abs/2307.16315

[23] L. Libkin, "Certain answers as objects and knowledge," *Artificial Intelligence*, vol. 232, pp. 1–19, 2016.

[24] F. Maia Polo, S. Maity, M. Yurochkin, M. Banerjee, and Y. Sun, "Weak supervision performance evaluation via partial identification," in *Advances in Neural Information Processing Systems*, vol. 37, 2024, pp. 135 704–135 740.

[25] L. Xia and V. Conitzer, "Determining possible and necessary winners under common voting rules given partial orders," *Journal of Artificial Intelligence Research*, vol. 41, pp. 25–67, 2011.

[26] T. Kuboyama, "How many labels does model choice need? certificates and budgets for selective prediction," 2026. [Online]. Available: https://arxiv.org/abs/2609.18622v1

[27] X. Wang, J. Tian, S. Lyu, Y. Gu, Z. Song, X.-W. Chang, and L. Ding, "Exact risk certificates for pairwise-judged reranking," 2026, arXiv:2610.24347. [Online]. Available: https://arxiv.org/abs/2610.24347

[28] L. Tan and C. L. A. Clarke, "A family of rank similarity measures based on maximized effectiveness difference," *IEEE Transactions on Knowledge and Data Engineering*, vol. 27, no. 11, pp. 2865–2877, 2015.

[29] A. A. Ageev and M. I. Sviridenko, "Pipage rounding: A new method of constructing algorithms with proven performance guarantee," *Journal of Combinatorial Optimization*, vol. 8, pp. 307–328, 2004.

[30] A. Srinivasan, "Distributions on level-sets with applications to approximation algorithms," in *Proceedings of the 42nd IEEE Symposium on Foundations of Computer Science*. IEEE, 2001, pp. 588–597.

[31] N. J. A. Harvey and N. Olver, "Pipage rounding, pessimistic estimators and matrix concentration," in *Proceedings of the Twenty-Fifth Annual ACM-SIAM Symposium on Discrete Algorithms*. Society for Industrial and Applied Mathematics, 2014, pp. 926–945.

[32] G. L. Nemhauser and L. E. Trotter, "Vertex packings: Structural properties and algorithms," *Mathematical Programming*, vol. 8, pp. 232–248, 1975.

[33] G. Katsirelos, "An analysis of core-guided maximum satisfiability solvers using linear programming," in *26th International Conference on Theory and Applications of Satisfiability Testing (SAT 2023)*, ser. Leibniz International Proceedings in Informatics (LIPIcs), vol. 271. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2023, pp. 12:1–12:19.

[34] M. Frances and A. Litman, "On covering problems of codes," *Theory of Computing Systems*, vol. 30, no. 2, pp. 113–119, 1997.